

Lab 1 - Practice guide

Images as numbers: arrays, raw pixel matching, nuisance variation, normalisation, and group results.

Submission: upload your executed Lab01_Light_Colour_Histograms_Student.ipynb by 23:59 on the lab day. Keep all figures and printed results visible.

How this notebook works

Notebook part	What you do
Part 1 - Common	Everyone runs the same template images and raw-pixel classifier. Complete the two inspection responses.
Part 2 - Personal	You enter your student ID. The notebook then creates your own scenario code, brightness conditions, shifts, group results, and intensity sweep.
Part 3 - Repair	You implement per-image normalisation and compare results on the same personal test set.
Part 4 - Stress test	You implement the assigned sweep and report the exact value and result produced for your scenario.

Before you start

Notebook part	What you do
Notebook	Open Lab01_Light_Colour_Histograms_Student.ipynb. Do not use the instructor notebook.
Student details	Complete your name, student ID, and lab date in the first markdown cell.
Running order	Run cells from top to bottom. Complete every cell marked Your task before using later results.
Written responses	Edit the markdown response areas. Quote values and scenario code from your own output.
Final run	Use Restart and Run All before submission. The completed notebook must run without errors.

Important: the first part is identical for everyone. Your personal work begins only after you replace DEMO-0000 with your student ID.

Part 1 - Common reference case

Complete these steps before entering your student ID.

Step 1 - Setup and labelled templates

Notebook part	What you do
Setup	Run the setup cell. It defines the display, accuracy, and helper functions.
Build a small labelled image set	Run the cell that creates train_images and train_labels. Keep the template grid and printed shape, dtype, minimum, and maximum visible.

Your task A1 - Inspect one image

Notebook part	What you do
Complete code	Complete describe_image. Return a dictionary containing shape, dtype, minimum, and maximum. Use the hints in the notebook.
Run A1	Run the cell again. Check that it prints information about a real template image.
Write A2	State the four properties you check when loading an image. State the template shape shown by your output.

Step 2 - Raw pixel template matching

Notebook part	What you do
Run	Run the mean absolute pixel distance classifier. It should give 100 percent accuracy on the unchanged template images.
Write A3	Explain why the classifier succeeds here. Include the alignment assumption: corresponding pixel locations must refer to the same scene parts.

Keep the evidence: the template grid, array properties, clean accuracy, and predicted labels must stay visible in the submitted notebook.

Part 2 - Your personal condition

This part changes when you enter your own student ID.

Step 3 - Generate your scenario

Notebook part	What you do
Edit one line	Replace DEMO-0000 with your own student ID. Do not change other code in that cell.
Run	Run the cell and record the printed scenario code, scenario name, two group conditions, and five sweep alpha values.
Check	If the notebook prints its DEMO warning, you have not entered your student ID correctly.

Step 4 - Build test images and measure failure

Notebook part	What you do
Run	Run the cell that builds test_images. Keep your personalised image grid, raw overall accuracy, and raw predicted labels visible.
Your task B1	Complete group_accuracy. It must return a percentage result for each group in test_groups. Use a Boolean mask for each unique group.
Write B2	Quote your scenario code, raw overall accuracy, raw Group A accuracy, raw Group B accuracy, and the condition of the worse group.

Your values will differ. Do not copy another student's reported numbers. Your scenario code and numerical evidence must match your own notebook output.

Parts 3-4 - Repair and stress test

Use the same personal test images throughout these final steps.

Your task C1 - Per-image normalisation

Notebook part	What you do
Complete normalise_one	Subtract the image mean and divide by its standard deviation. Keep the safe constant-image case requested in the code comments.
Complete normalise_batch	Apply normalise_one to every image and stack the results.
Run C1	Check the printed mean and standard deviation. For a non-constant image they should be near 0 and 1.
Evaluate repair	Run the repair cell. It reports overall accuracy and group scores on the same personal test set.
Write D1	Use your scenario code and at least three exact values. Explain what improved, what remained, and whether a shift or clipping can be repaired by intensity normalisation.

Your task E1 - Personalised intensity sweep

Notebook part	What you do
Complete code	Complete evaluate_personal_sweep. For each assigned alpha, transform every training image with sweep_shift, run raw-pixel prediction, and store accuracy in a dictionary.
Run E1	Run the cell using your five assigned sweep values.
Write E2	Report scenario code, lowest-accuracy alpha, its accuracy, and whether the result supports the claim that raw pixel distance is nuisance-sensitive. Name the signal and the nuisance.

Submit by 23:59 on the lab day

Notebook part	What you do
Complete	A1, A2, A3, B1, B2, C1, D1, E1, and E2 are complete.
Personal evidence	Your own student ID, scenario code, image grid, group scores, repair result, and sweep output are visible.
Final check	Restart and run all cells without errors. Upload the executed student notebook.